

4.4. Camber Line Design: The Uniform Load Distribution

It is possible to use thin airfoil theory to design an airfoil having a desired load distribution. In the previous section, we assumed that the shape of the camber line was known. We used this known shape to evaluate the vortex strength distribution necessary to make the camber line a streamline of the flow. We could, however, treat the vortex strength distribution as a known and solve for the shape of the camber line. Because the lift per unit area at any axial position along the airfoil chord is proportional to the vortex strength at that point, a known vortex strength distribution is equivalent to a known chordwise lift distribution. As an example of how one would design an airfoil for a desired load distribution, we shall consider the case of designing an airfoil for uniform load.

For a uniform chordwise load distribution, the value of the vortex strength, γ , in Eq. (4.3.12) is constant over the chord. Thus, Eq. (4.3.12) can be written as

$$\frac{\gamma}{2\pi} \int_{x_o=0}^c \frac{dx_o}{x-x_o} = V_\infty \left(\alpha - \frac{dy_c}{dx} \right) \quad (4.4.1)$$

or

$$\frac{dy_c}{dx} = \alpha + \frac{\gamma}{2\pi V_\infty} \int_{x_o=0}^c \frac{dx_o}{x_o - x} \quad (4.4.2)$$

The integral on the right hand side of Eq. (4.4.2) is singular at the point $x_o = x$. However, it can be evaluated using

$$\begin{aligned} \int_{x_o=0}^c \frac{dx_o}{x_o - x} &= \lim_{\varepsilon \rightarrow 0} \left[\int_{x_o=0}^{x-\varepsilon} \frac{-dx_o}{x-x_o} + \int_{x_o=x+\varepsilon}^c \frac{dx_o}{x_o - x} \right] \\ &= \lim_{\varepsilon \rightarrow 0} [\ln(x-x_o)_{x_o=0}^{x-\varepsilon} + \ln(x_o-x)_{x_o=x+\varepsilon}^c] \\ &= \lim_{\varepsilon \rightarrow 0} [\ln(\varepsilon) - \ln(x) + \ln(c-x) - \ln(\varepsilon)] \\ &= \ln(c-x) - \ln(x) \\ &= \ln\left(\frac{c-x}{x}\right) \end{aligned} \quad (4.4.3)$$

Using Eq. (4.4.3) in Eq. (4.4.2) gives

$$\frac{dy_c}{dx} = \alpha + \frac{\gamma}{2\pi V_\infty} \ln\left(\frac{c-x}{x}\right) \quad (4.4.4)$$

From the Kutta-Joukowski law, the section lift for an airfoil with constant vortex strength is

$$\tilde{L} = \rho V_\infty \Gamma = \rho V_\infty \int_{x_o=0}^c \gamma dx_o = \rho V_\infty \gamma \int_{x_o=0}^c dx_o = \rho V_\infty \gamma c \quad (4.4.5)$$

So the section lift coefficient can be written

$$\tilde{C}_L \equiv \frac{\tilde{L}}{\frac{1}{2} \rho V_\infty^2 c} = \frac{2\gamma}{V_\infty} \quad (4.4.6)$$

or

$$\frac{\gamma}{V_\infty} = \frac{\tilde{C}_L}{2} \quad (4.4.7)$$

Using Eq. (4.4.7) in Eq. (4.4.4), we have

$$\frac{dy_c}{dx} = \alpha + \frac{\tilde{C}_L}{4\pi} \ln\left(\frac{c-x}{x}\right) = \alpha + \frac{\tilde{C}_L}{4\pi} [\ln(c-x) - \ln(x)] \quad (4.4.8)$$

Integrating Eq. (4.4.8) gives

$$y_c = \alpha x - \frac{\tilde{C}_L c}{4\pi} \left[\left(1 - \frac{x}{c}\right) \ln\left(1 - \frac{x}{c}\right) + \left(\frac{x}{c}\right) \ln\left(\frac{x}{c}\right) - 1 \right] + C \quad (4.4.9)$$

where C is a constant of integration. By definition, y_c is zero at both $x = 0$ and $x = c$,

$$y_c(0) = 0 \quad (4.4.10)$$

$$y_c(c) = 0 \quad (4.4.11)$$

From Eq. (4.4.10) we must have

$$C = -\frac{\tilde{C}_L c}{4\pi} \quad (4.4.12)$$

In view Eq. (4.4.12), Eq. (4.4.11) requires

$$\alpha = 0 \quad (4.4.13)$$

This means that a uniform load distribution is only possible at zero angle of attack. However, at zero angle of attack, we can design an airfoil to have a **uniform load distribution** and any desired lift coefficient by constructing the camber line according to the equation,

$$\frac{y_c}{c} = -\frac{\tilde{C}_{Ld}}{4\pi} \left[\left(1 - \frac{x}{c}\right) \ln\left(1 - \frac{x}{c}\right) + \left(\frac{x}{c}\right) \ln\left(\frac{x}{c}\right) \right] \quad (4.4.14)$$

where $\tilde{C}_{Ld}$ is the design section lift coefficient for zero angle of attack.

The camber line geometry given by Eq. (4.4.14) results in minimum induced velocity for a given design lift coefficient and thickness distribution. Many airfoil sections have been designed using this camber line. The best known of these is the NACA 1-series family of airfoil sections. This series represents the first attempt to develop airfoil sections having a desired pressure distribution and it was the first of seven NACA airfoil families to be designed using theoretical methods.

To obtain some insight into the importance of camber line design, consider the airfoil comparison that is shown in Fig. 4.4.1. Both airfoil sections shown in this figure have about the same maximum camber and thickness distribution. The difference between these two airfoils is that the section on the left is a standard NACA 4412, while the airfoil section on the right has a similar thickness distribution and maximum camber but has been modified to have the uniform load camber line that is given by Eq. (4.4.14). These two airfoil sections look very much the same. The only visible difference is that the uniform load airfoil seems to carry its camber further back toward the trailing edge. The pressure distributions for these two airfoils, however, are quite different. The section on the right displays a pressure difference between the upper and lower surface that is nearly constant from the leading edge to the trailing edge. This is exactly the condition for which the camber line was designed. The NACA 4412 airfoil, on the other hand, displays a pressure difference between the upper and lower surface that varies considerably between the leading and trailing edge. In fact, in the region very near the leading edge of the NACA 4412, the local lift is actually negative at zero angle of attack. This unnecessarily reduces the total section lift coefficient and makes pitching moment more negative.

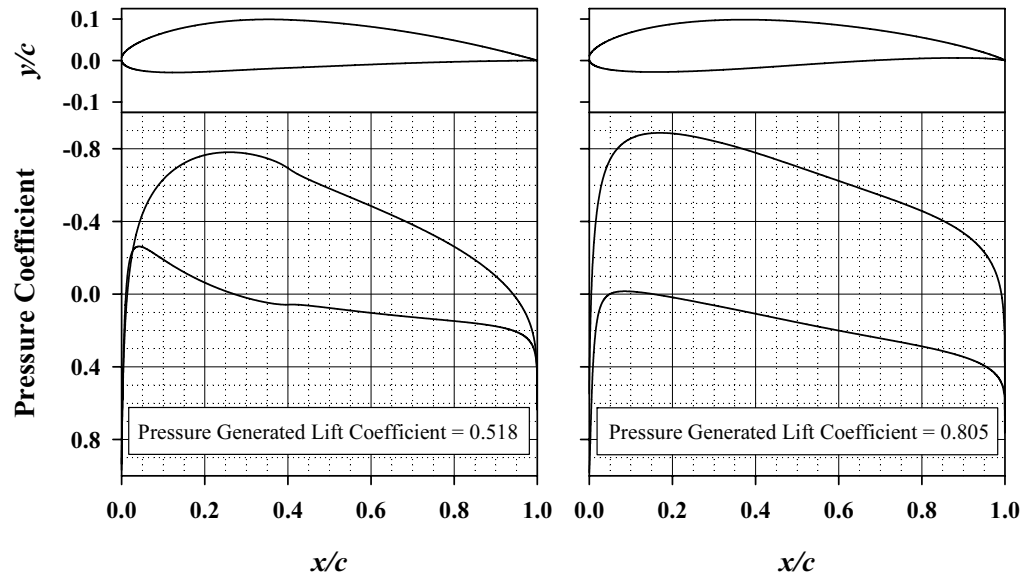


Figure 4.4.1. The zero angle of attack pressure distribution for an airfoil with a camber line designed for uniform load (right), compared with that for a similar NACA 4-digit series airfoil (NACA 4412 left).

The boundary layer on the surface of an airfoil section, and hence the section drag coefficient, are profoundly affected by the surface pressure distribution. The pressure distribution around the surface of any airfoil section depends on both the thickness distribution and the shape of the camber line. The pressure difference between the upper and lower surface, at zero angle of attack, depends primarily on the shape of the camber line, while the mean pressure between the upper and lower surface is controlled mostly by the thickness distribution. Since we do not have an analytical method for obtaining a thickness distribution to give a desired mean pressure distribution, the process of designing a thickness distribution to achieve a desired surface pressure distribution with a given camber line geometry becomes an iterative process. This is, however, an important part of the airfoil design process.

Notice that, on both the upper and lower surfaces of the uniform load airfoil shown in Fig. 4.4.1, the pressure is increasing with x over most of the surface. A pressure distribution that increases in the direction of flow is called an adverse pressure gradient because of its adverse effects on the boundary layer. An adverse pressure gradient causes the boundary layer to become thicker and more likely to separate, producing higher drag and reducing lift. An adverse pressure gradient also promotes boundary layer instability, causing the boundary layer to become turbulent closer to the leading edge. A turbulent boundary layer produces more drag than a laminar boundary layer but is less likely to separate under the influence of an adverse pressure gradient. Thus, to minimize drag and decrease the possibility of boundary layer separation, we may wish to design an airfoil section so as to minimize adverse pressure gradients on the airfoil surface.

The pressure gradient on an airfoil surface can be controlled through the design of the camber line and the thickness distribution. To design an airfoil section for a desired pressure distribution at some design lift coefficient, the camber line can be designed, using thin airfoil theory, to give the desired pressure difference between the upper and lower surfaces, at some design angle of attack. A thickness distribution can then be designed in an iterative manner, using a panel code, to give the desired mean pressure distribution when used with a symmetric airfoil section at zero angle of attack. When this camber line and thickness distribution are combined, the resulting airfoil section will have very nearly the desired pressure distribution, at the design angle of attack. Although this method of superposition has little theoretical justification, it seems to work quite well for most low Mach number applications. For example, the thickness distribution shown in Fig. 4.4.2 was designed to give a nearly constant pressure over the leading 60 percent of the symmetric airfoil section at zero angle of attack. When this thickness distribution is combined with the same uniform load camber line that is used in the airfoil on the right-hand side of Fig. 4.4.1, we obtain the airfoil shown on the right in Fig. 4.4.2. Notice that the cambered section displays nearly the same mean pressure distribution as the symmetric section at zero angle of attack.

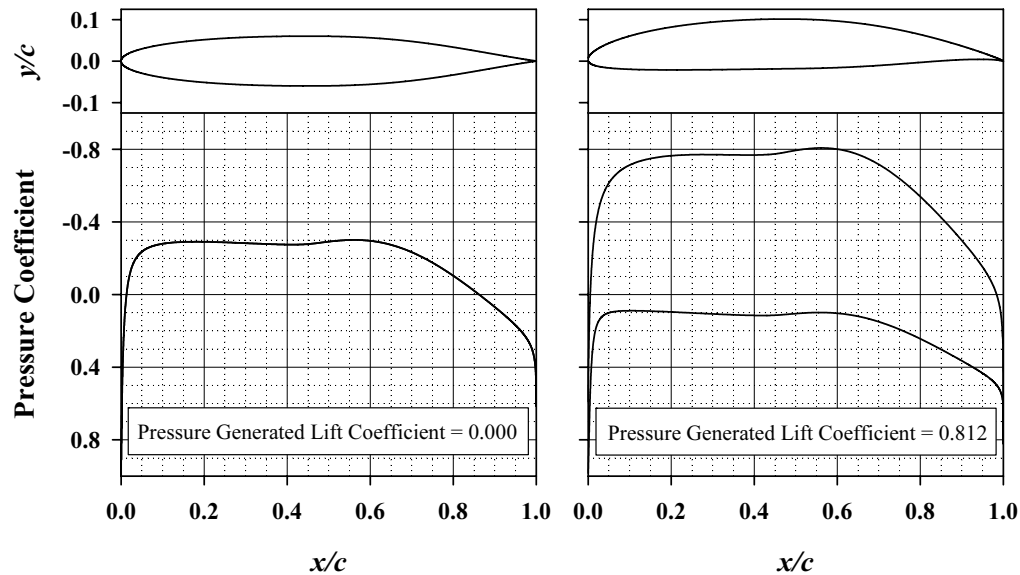


Figure 4.4.2. The zero angle of attack pressure distribution for an airfoil section with a camber line designed for uniform load (right), combined with a thickness distribution designed for a nearly constant mean pressure over the leading 60% of the airfoil section (left).

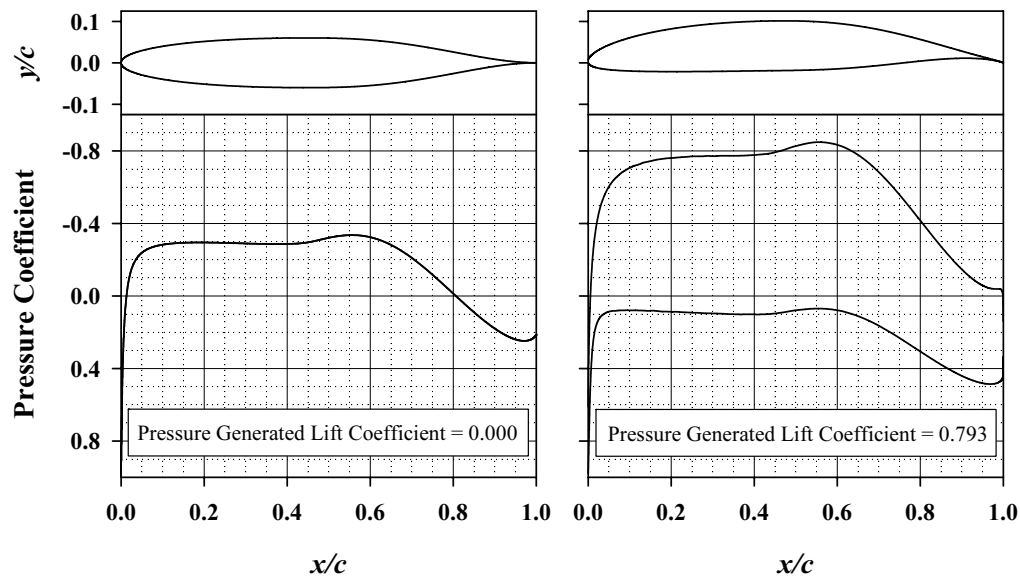


Figure 4.4.3. The zero angle of attack pressure distribution for an airfoil section with a camber line designed for uniform load (right), combined with a thickness distribution having a zero trailing edge angle (left).

Notice that very close to the trailing edge of the airfoils shown in Figs. 4.4.1 and 4.4.2 there is a small region with a very large adverse pressure gradient. This almost always causes a small separation near the trailing edge. This slight separation has little effect on the section lift, but does add somewhat to the section drag. Using a thickness distribution with a trailing edge angle of zero will minimize this effect. A thickness distribution of this type, combined with a uniform load camber line, is shown in Fig. 4.4.3. Notice that, here again, the cambered section displays nearly the same mean pressure distribution as the symmetric section at zero angle of attack. Manufacturing and durability issues often preclude the use of a zero trailing edge angle.

When the camber line geometry given by Eq. (4.4.14) is used with any reasonable thickness distribution, the pressure difference between the upper and lower surfaces, at zero angle of attack, will be nearly constant. This gives the uniform chordwise load distribution for which the camber line was designed. Following a procedure similar to that used to obtain Eq. (4.4.14), a camber line can be designed for other load distributions as well. When using this process to design a camber line for a given load distribution, it is always found that the given load distribution can be achieved only at one particular angle of attack. This angle of attack is called the “ideal angle of attack” for that particular camber line. The section lift coefficient corresponding to this ideal angle of attack is called the “ideal” or “design” lift coefficient. The ideal angle of attack for an airfoil constructed with a uniform load camber line is zero.

Useful camber lines have been designed to give a load distribution that is uniform between $x/c = 0$ and $x/c = a$ and that linearly decreases from this point to zero at $x/c = 1$. This load distribution results in the camber line geometry

$$\begin{aligned} \frac{y_c}{c} = \frac{\tilde{C}_{Ld}}{2\pi(1+a)} \left\{ \frac{1}{1-a} \left[\frac{1}{2} \left(a - \frac{x}{c} \right)^2 \ln \left| a - \frac{x}{c} \right| - \frac{1}{2} \left(1 - \frac{x}{c} \right)^2 \ln \left(1 - \frac{x}{c} \right) \right. \right. \\ \left. \left. + \frac{1}{4} \left(1 - \frac{x}{c} \right)^2 - \frac{1}{4} \left(a - \frac{x}{c} \right)^2 \right] - \left(\frac{x}{c} \right) \ln \left(\frac{x}{c} \right) - g - h \frac{x}{c} \right\} \end{aligned} \quad (4.4.15)$$

where

$$g = \frac{1}{1-a} \left[a^2 \left(\frac{1}{2} \ln a - \frac{1}{4} \right) + \frac{1}{4} \right] \quad (4.4.16)$$

$$h = \frac{1-a}{4} [2 \ln(1-a) - 1] - g \quad (4.4.17)$$

Here again, $\tilde{C}_{Ld}$ is the design section lift coefficient at the ideal angle of attack, which for this family of camber lines is given by

$$\alpha_i = \frac{\tilde{C}_{Ld} h}{2\pi(1+a)} \quad (4.4.18)$$

The NACA 6-series family of airfoil sections uses this camber line geometry. For more details on this family of airfoils see Abbott and Von Doenhoff (1949).